

Analyzing Sentiment and Emotion in Song Lyrics

Evaluating VADER and DistilBERT on Lyrical Text

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Sentiment analysis is a popular utility of Natural Language Processing. It is commonly used to analyze customer feedback for products and services. A unique application would be evaluating the content of entertainment, such as music. Can Natural Language Processing tools accurately analyze sentiment and emotion from song lyrics?

The objectives of this project were: 1) to obtain a dataset of songs representative of popular music; 2) to optimize the dataset for processing; 3) to utilize a model to classify sentiment in song lyrics; 4) to utilize a model to classify emotion in song lyrics; 5) to run the project pipeline for overall dataset analysis; 6) to generate playlists from the dataset; and 7) to evaluate performance on overall sentiment and specific song lyrics for two different artists.

Gemini AI was used for some code generation (e.g. data visualization, artist sentiment search, playlist output) and problem-solving (e.g. preventing over-fitting).

1. The Dataset

This project used the “Spotify Lyrics Dataset” by Eva Bot on Kaggle (2023). It contained data captured from Spotify and lyrics web pages. It contained columns for recording artist(s), song title, song genre(s), lyrics, explicit content tag, Spotify artist ID, and Spotify song ID. It featured over 300 unique artists representing popular music spanning decades of recorded music. Names varied from A\$AP Rocky to Tony Bennett, and genres spanned from heavy metal to dance.

2. Optimizing the Dataset

First, song entries with a null value were removed. Second, song entries with multiple contributing artists had to be standardized with one artist. The first artist named was extracted and the rest discarded. Third, genres had to be standardized. The first genre listed for each song was extracted and the rest were discarded. This resulted in 137 unique genres. Genres with under 40 instances in the list were removed, clearing out 83 niche genres from the dataset. The remaining 54 genres were filtered into 13 umbrella genres. For example: “gangster rap” would be filtered into “rap,” “modern blues” was filtered into “blues”, etc.

Finally, lyric length was standardized with a minimum of 20 tokens. The VADER sentiment analysis used in this project was developed with a specialization for analyzing short social media posts and customer reviews (Hutto & Gilbert, 2014, pp. 1). The DistilBERT emotion analysis was also trained on a Twitter data set (Savani, 2021). Therefore, it was reasoned that a 20-token minimum would be sufficient for sentiment analysis with VADER.

3. Model for Classifying Sentiment

VADER (Valence Aware Dictionary and sEntiment Reasoner) is a rule-based sentiment analysis model developed by C.J. Hutto and Eric Gilbert in 2014 (pp. 1). It was chosen for its proficiency with analyzing informal speech from social media and customer review datasets. Lyrics in popular music often utilize informal speech, strong emotional expression, and repetition. It was reasoned that lyrics were similar enough to the datasets which VADER was demonstrated to analyze well. Also, its computational efficiency and lack of need for training made it easy and quick to use.

VADER boasted a lexicon of over 7,500 features from well-established word banks and their own research (Hutto & Gilbert, 2014, pp. 5). It was computationally light, did not hide its methodology in a black box, and eschewed training in favor of rule-based modeling (pp. 9). Notably, it out-performed human beings who were tasked with analyzing sentiment to verify the model's conclusions (pp. 1).

VADER's five conditions for sentiment analysis were as follows: 1) presence of punctuation (specifically exclamation points); 2) using fully capitalized words; 3) degree adverbs ("very," "quite", etc.); 4) contrastive conjunction "but" ("this was great but that was awful"); and 5) using a trigram to check for negating words before a phrase ("The food here isn't really all that great") (Hutto & Gilbert, 2014, pp. 6).

4. Model for Classifying Emotion

Bhadresh Savani's "base-uncased-emotion" pre-trained DistilBERT model was selected for emotion analysis. It is a smaller implementation of HuggingFace's BERT (Bidirectional Encoder Representations from Transformers) language model and was trained for emotion analysis specifically with the Twitter-Sentiment-Analysis dataset. (Savani, 2021). It features six emotional classifications: joy, anger, sadness, fear, love, and surprise.

A DistilBERT model was chosen for emotion classification due to its efficiency and power. The base DistilBERT transformer was developed by Sanh, Debut, Chaumond, and Wolf in 2019 (pp. 1). It was 40% smaller, 60% faster, and has 97% of its parent model's language comprehension. It features BERT's bidirectional encoder-based transformer architecture (Kumar, 2024).

5. Project Pipeline

Sentiment analysis with VADER was performed on the optimized dataset. Each song was assigned a floating point value denoting sentiment between -1 and 1 in a new column. Statistical description of the weights revealed that the mean was 0.19, denoting a slightly positive sentiment across the data set. The standard deviation was 0.88, showing a wide spread in sentiment scores. The minimum and maximum were at ± 0.99 , the extremes in both domains. The dataset was overwhelmingly determined to be positive. Scores between the 1st and 2nd quartiles ranged from -0.93 to 0.77, demonstrating a massive increase across only 25% of the entries. Finally, 25% of the entries were above 0.98.

An 80% train and 20% test split were created. Discrete classes for turning sentiment scores into labels were established: a range of -1 to -0.051 was classified as negative, a range of -0.05 to 0.05 as neutral, and 0.051 to 1.0 as positive. A TF-IDF vectorizer was used to generate numerical representations for machine analysis. English stop-words were removed, the top 5,000 words were selected, and only unigrams were used.

Logistic Regression used the training set to learn the relationship between the sentiments derived from VADER and the TF-IDF vectors. A balanced class weight was chosen because of the high proportion of positive entries in the dataset observed in the statistical description. L2 regularization with a penalty term and C value of 0.1 were applied with the goal of reducing overfitting. It used that training to classify the sentiments for the test set and output a confusion matrix with the results.

The confusion matrix output demonstrated good performance for identifying negative and positive entries. Positive entries enjoyed the highest marks (83% precision, 81% recall, and 82%

F1-Score). Negative entries were slightly lower but still good (74% precision, 74% recall, and 74% F1-Score). Positive scores may have been higher due to the higher number of positive entries in the test dataset. Finally, neutral scores were the lowest (27% precision, 73% recall, 39% F1-score). This may be a result of only 11 instances of neutral entries being in the test dataset.

The macro average is at 65%, revealing a hit taken from poor performance with identifying neutral lyrics. The weighted average is significantly higher at 79%, showing that the model performs fairly well overall. Training and test accuracy were shown to be at 82.8% and 78.6%, making a 4.2% gap. The model slightly avoids overfitting.

Next, the continuous sentiment scores were filtered into discrete values in a new column. This provided nuanced sentiment analysis for review with the upcoming emotion analysis. It would be more helpful to determine the model's correctness with degrees of intensity (e.g., "slightly negative" classification on a song which seems "slightly positive" is more descriptive than the negative-neutral-positive trinary used earlier).

The neutral domain was retained at -0.05 and 0.05 . The positive and negative sentiments were graded in increasing degrees of slightly, standard, very, or extremely positive / negative in increments of approximately ± 0.2375 (-0.051 to -0.28751 for Slightly Negative, -0.5251 to -0.28751 for Negative, etc.). For reference purposes, counts were performed on the sentiment categories to get overall amount of positive, negative, and neutral results (4482, 2989, and 54, respectively).

Several visual outputs were generated for ease of viewing. A table and bar graph showing average compound sentiment score by genre were generated. Adult standards, country, and blues

were shown as being the most positive genres, while metal, hip hop, and rap were shown as being the most negative. Bar graphs for top and bottom artists by compound sentiment were output, showing Clean Bandit and DING DONG as most positive and Tony Yayo and D12 as most negative. It was peculiar that country and blues ranked with more positive sentiments than “pop” or “dance” music, as those genres are conventionally known as dealing with life struggles. This may have hinted at poor sentiment analysis by VADER.

Savani’s DistilBERT model was loaded. Lyrics were truncated to DistilBERT’s limit of 512 tokens. The lyrics column was analyzed to extract emotion and then assigned to a new column. The approximate emotion percentages showed 37% were tagged with “joy,” 25% for both “anger” and “sadness,” 6% for “fear,” 5% for “love,” and 1% for “surprise.”

Sentiment tallies did not match squarely with the emotion tallies. Approximately 60% of the entries were labeled as “positive” by VADER, whereas only 37% were labeled with “joy” by DistilBERT. Added with other emotions which can be positive, such as “love” and “surprise,” there would still be 1,280 more songs to apply emotions to. This meant that they would have negative emotions assigned to them, such as “anger,” “sadness,” or “fear.” There would be contradictory sentiment and emotion pairings (e.g. “positive” sentiment with “anger” emotion). The models’ different implementations would account for this discrepancy.

Emotion data was output as charts. One chart shows emotion frequency. Another chart shows a stacked bar chart with emotion frequency by genre. That chart provided the most interesting information. Metal, rap, and hip hop showed the highest proportions for anger. Country showed the highest frequency of fear, while folk showed the highest frequency of sadness. Dance, blues, and punk did not have any entries for surprise.

6. Playlist Generation

The processed dataset for emotion may be used to generate a playlist. The playlist may be based on emotion or genre and emotion. Each playlist is generated randomly on command. A csv file is output containing playlist order, Spotify song ID, song, artist, genre, a compound sentiment score, and emotion. The song ID column should be copied and pasted into the playlist screen in the Spotify desktop program. The songs will automatically be queued into the player for playback.

The results of the playlist may vary due to the NLP pipeline's difficulty with analyzing emotion. For example, songs analyzed as being joyful by Savani's DistilBERT model may not actually contain joyful lyrics.

7. Artist and Song Analysis

A search function for retrieving an artist info in the dataset was implemented. The header table showed song count, predicted sentiment accuracy, compound sentiment score, actual sentiment, and predicted sentiment. The next table was populated a row for each song, its genre, the sentiment degree, and emotion. This provided a quick way to look up an artist and the pipeline's analysis of their content. It was used to select songs for NLP labelling analysis.

"Lust for Life" by Iggy Pop was the first song selected. Its was labeled as "Extremely Positive" with a "joy" emotion. Surprisingly, its lyrical content addressed struggling with addictions (alcohol, drugs, etc.), disillusionment with love, and difficulties from bad decisions (Pop, 1977). The lyrics were closer to a negative sentiment with sad emotion. The models failed to pick up the irony and dark humor of the song. The chorus, "I got a lust for life," would be taken at face-value by models not trained to detect sarcasm, swaying analysis.

“Gold Rush” by Taylor Swift was another song labeled as “Extremely Positive” with “joy.” The lyrics for the song were more straightforward than Pop’s “Lust for Life,” plainly describing disillusionment with a competitive relationship (“Everybody wants you, But I don't like a gold rush”) and self-esteem issues (“What must it be like to grow up that beautiful?... I can't dare to dream about you anymore”) (Swift, 2020). The sentiment was negative and emotion leaned towards anger. The chorus featured the phrase “I don’t like” prominently, but VADER did not register it as a negative sentiment. Surprisingly, Savani’s DistilBERT model did not take context for emotion from the surrounding words (“I don't like that anyone would die to feel your touch... I don't like that falling feels like flying 'til the bone crush”). The presence of “gold” in the chorus may influence both models to esteem the song as positive. The word is often used as a positive indicator in speech (“This is the gold standard,” or “it was worth its weight in gold”). Their training on customer reviews may have biased them if the word was used in a positive context.

The pipeline’s analysis was investigated on songs labeled as negative. “Corruption” by Iggy Pop had a “Very Negative” sentiment and “anger” emotion. It featured lyrics decrying the corruption found in various aspects of life (Pop et al., 1999). The sentiment matched the lyrics well (“From the fact, fact, fact of the soulless, To the pact, pact, pact with hell”). It did not seem to match the emotion label of “anger” so well. Instead, it leaned closer to a “sadness” or “fear” sentiment (“Corruption... Rules my soul, Corruption... Chills my bones”).

Next, “Lavender Haze” by Taylor Swift was inspected, labeled as “Extremely Negative” sentiment with “fear” emotion. The lyrics described Swift’s emotions over public scrutiny from her fame. She plainly described her sadness (“And you don’t really read into my melancholia”) and cynicism (“I’m damned if I do give a damn what people say”). The extreme sentiment score

may be a result of the profanity which Swift uses in the song. Fear was a decent label for the emotion based on the topic of public scrutiny, although she expressed defiance at the prospect of letting it rule her life (“No deal, The 1950s shit they want from me”). “Anger” or “sadness” may have been more accurate emotion labels due to that defiance or her aforementioned melancholia.

Compound sentiment analysis based on Pop’s 35 songs in the dataset was positive with a compound sentiment score of 0.355. This was in contrast to Encyclopedia.com’s entry on him, highlighting his continuous use of confrontational and pessimistic themes (n.d.).

Compound sentiment analysis based on Swift's 18 songs in the dataset was slightly positive with a compound sentiment score of 0.105. Ruggiero et al. found through a Natural Language Processing study that Swift’s early career was positive, but had incorporated more neutral and negative sentiments time progressed (2024). Those findings seemed to be reflected in the compound score.

8. Conclusion

The VADER and Savani DistilBERT models were only partially effective at discerning sentiment and emotion based on the analysis of Pop and Swift’s songs. Both models could not discern the sarcasm from Pop or Swift in their negative songs which used sarcastically positive language. The models fared better when analyzing negative sentiment and emotions. Still, the DistilBERT model had difficulty recognizing what sentiment was truly being communicated with both artists (“anger” vs. “sadness” or “fear”, “fear” vs. “anger” or “sadness”).

The discrepancies between content and analysis seemed to result from the way the models were implemented. VADER’s lexicon and condition-based classification struggled with the nuance of lyrics. DistilBERT’s training with social media content did not prepare it for the

challenge, either. There was a unique linguistic complexity found in song lyrics which did not overlap with the models' respective domains.

A study carried out by Mara et. al. highlighted some problems VADER had with capturing sentiment in reviews, concluding that it “may fail to account for subtle contextual factors” (2025, pp. 17). They also noted that its sentiment scores failed to correlate with review ratings (pp. 18). The discrepancy between VADER and DistilBERT’s analysis of the song lyric dataset may be another incidence of that behavior. The analyzed sentiment does not match with the “rating,” or feelings being expressed by the song lyrics. VADER says “positive,” but the lyrics state otherwise.

Mara et. al. examined the difference between VADER and BERT, the parent model of DistilBERT (2025, pp. 4, 17). They stated: “VADER's lexicon-based approach can oversimplify context, while BERT’s transformer-based architecture captures nuanced expressions but may amplify biases inherent in training data.” DistilBERT’s biases may have been shown with its analysis of Swift’s “Gold Rush,” where “gold” was interpreted in a positive context rather than a negative one. Even its bidirectional capability and layered processing could not override its training, ignoring the “I don’t like” placed right before “a gold rush.”

The project may be improved in several ways. It can be implemented with models specifically trained on song lyrics. Pre-existing models can also be fine-tuned to analyze song lyrics. More cutting-edge models may be used as well. VADER was released in 2014 and DistilBERT was released in 2019. Newer technology may perform better than models which are over seven years old.

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